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2 State of the Art

This chapter reviews the state of the art in appearance-based gaze estimation with standard RGB cameras. Starting from the methodological foundations (Section 2.1), it describes the specific conditions of webcam-based systems (Section 2.2), the most important datasets (Section 2.3), and the network architectures (Section 2.4) and pretrained models (Section 2.5) in use. It then discusses approaches to personalization, domain shift, test-time adaptation, and data quality before deriving the research gaps addressed by the present work.

2.1 Foundations and Evolution of Appearance-Based Gaze Estimation

Eye tracking aims to determine where a person is looking from image information of the eyes or face. Depending on the application, either the three-dimensional gaze direction, typically expressed as a pair of pitch and yaw angles, or the two-dimensional *Point of Regard* (PoR) is estimated, that is, the point on a screen at which the person is looking. The present work focuses on 2D PoR estimation from images of a commodity webcam.

2.1.1 Classical Appearance-Based Methods

Appearance-based methods learn a direct mapping from image intensities to gaze targets without fitting an explicit geometric eye model. Early approaches combined hand-crafted image features such as Local Binary Patterns (LBP), Histograms of Oriented Gradients (HoG), or SIFT descriptors with linear regressors, random forests, or support vector regression (SVR). These methods are computationally inexpensive, but their accuracy degrades markedly under head movement and changing illumination, because hand-crafted features separate these variations only insufficiently from the actual gaze information.

2.1.2 Pioneering Deep Learning Approaches

The advent of convolutional neural networks (CNNs) enabled an end-to-end mapping from face and eye images to screen positions. Krafka et al. [1] introduced *iTracker*, one of the first CNN architectures for mobile eye tracking without additional hardware. The model

processes the left eye, the right eye, the face, and a binary *face grid* in separate AlexNet-like streams and fuses the resulting features to predict the 2D gaze position. Without calibration, iTracker achieves a prediction error of 1.71 cm on smartphones and 2.53 cm on tablets. For personalization, a user-specific SVR is trained on the features of layer `fc1`, using recordings from up to 13 fixed calibration locations. With all 13 points, the error decreases to 1.34 cm and 2.12 cm, respectively; with very few calibration points, however, the SVR may overfit. The model runs in real time on mobile devices.

A major contribution of this work was the release of the GazeCapture dataset (Section 2.3), which was collected via crowdsourcing on smartphones and tablets under in-the-wild conditions and still serves as a benchmark and as a source of pretraining data.

Relevance to This Work. iTracker established three elements that remain relevant today: (1) the GazeCapture dataset, (2) personalization through calibration data, and (3) the fusion of features from multiple image regions. The present work uses the same data foundation but replaces the AlexNet/SVR pipeline with a ResNet18 backbone [2] and Smooth L1 regression. In addition, it adopts a subject-disjoint evaluation protocol (`norm_subject`) instead of a random split of individual frames, and it addresses label noise already during data acquisition through explicit logging of the acquisition phase (`WAIT` vs. `MOVE`).

2.1.3 Taxonomy of Gaze Estimation Methods

Gaze estimation methods can generally be divided into three methodological classes:

1. **Model-based (geometric) methods** use explicit 3D eyeball models, pupil ellipse fitting, and corneal reflections. They usually require high-resolution images under infrared illumination and are error-prone under natural ambient light.
2. **Feature-based methods** extract geometric facial and ocular features, such as eye corners or iris centers, and compute gaze vectors from them.
3. **Appearance-based methods** use the eye or face image itself as input and directly regress 2D coordinates or 3D gaze angles with deep neural networks.

Owing to advances in deep learning, appearance-based methods have become the dominant paradigm under unconstrained conditions.

Cheng et al. [3] likewise distinguish three groups of remote-camera methods: methods based on a recovered 3D eye model, regression methods based on 2D eye features, and appearance-based methods. They organize deep learning approaches from four perspectives: feature extraction, network architecture, personal calibration, and devices and platforms. A major contribution is a benchmark that unifies pre-processing (face and eye detection, data rectification) and post-processing (conversion between 2D gaze points and 3D gaze

vectors), thereby enabling fair comparisons between methods. According to the benchmark, appearance-based methods typically achieve about 5° on datasets with a short user-camera distance or high-resolution cameras, whereas considerably larger errors occur on datasets with a large distance and low resolution, such as Gaze360 and RT-GENE. As future research directions, the authors name (1) the extraction of more robust, environment-invariant gaze features, (2) high accuracy with fast and simple calibration, noting a trade-off between calibration time and accuracy, and (3) the interpretability of learned features.

Further surveys provide additional historical and methodological context. Liu et al. [4] trace the development from classical to deep learning-based methods and serve primarily as a historical reference in this work. Pathirana et al. [5] focus on deep learning-based approaches for single-user and multi-user scenarios and compare CNN-based architectures, some of which use ImageNet-pretrained backbones. However, more recent pretrained representations such as DINOv2, CLIP, or MAE-based models are not the focus of these surveys; for benchmark-based comparisons, Cheng et al. [3] therefore serves as the main reference.

2.2 Webcam-Based Eye Tracking: In-the-Wild vs. Laboratory Settings

Compared with dedicated eye-tracking hardware based on infrared illumination, webcam-based systems offer low cost and high availability, as they can be integrated into laptops, desktop computers, and mobile devices without additional hardware. This advantage comes at the price of considerably more difficult operating conditions:

- **Infrared-based systems** use high-resolution cameras, infrared light sources, and corneal reflections. Liu et al. [4] report accuracies of typically 0.5°–2° for 2D mapping-based and 3D model-based methods. Commercial remote trackers operate at specified distances (e.g., 50 cm to 80 cm) and achieve accuracies of 0.3°–0.5° under optimal conditions.
- **Webcams under in-the-wild conditions** provide images with lower resolution and without controlled illumination. In addition, ambient lighting fluctuates, the geometry between camera and screen varies, and distances and head poses are unconstrained. For appearance-based methods, Liu et al. [4] report accuracies typically above 3°; the benchmark by Cheng et al. [3] shows typical errors of about 5° at short distances and considerably larger errors at large distances and low resolution.

Two influential works nevertheless demonstrate that webcam-based eye tracking is practically usable. Papoutsaki et al. [6] present *WebGazer*, an open-source eye tracker that runs entirely in the browser. Instead of an explicit calibration, WebGazer continuously learns a mapping from eye features to screen positions from user interactions such as mouse clicks and

cursor movements, using ridge regression. The achieved accuracy suffices for a coarse estimate of the attended region, but not for fine-grained analyses. Valliappan et al. [7], in contrast, show that a CNN pretrained on GazeCapture, fine-tuned with calibration data and complemented by an additional person-specific regression model, achieves accuracies of 0.6°–1° on smartphones at a viewing distance of 25 cm to 40 cm. Personalization with about 30 seconds of calibration data reduces the error from 1.92 cm to 0.46 cm. According to the authors, this accuracy is comparable to that of considerably more expensive mobile eye trackers, although it was achieved under favorable conditions (near-frontal head pose, short distance). Both works underline that personalization is the decisive factor for the accuracy of camera-based systems without dedicated hardware.

Beyond these technical factors, user-specific characteristics such as eye shape, facial geometry, eyeglasses, and habitual head pose affect accuracy. Generic models therefore often exhibit a systematic offset for new users, which makes personalization a central research topic (Section 2.6). The present work investigates precisely these conditions, including different operating systems (Ubuntu and Windows), eyeglasses, and backgrounds.

2.3 Large-Scale Gaze Datasets

Modern appearance-based methods rely on large and diverse datasets. Table 2.1 summarizes the datasets relevant to this work.

- **MPIIGaze** [8] is one of the first standardized in-the-wild datasets. It comprises 213 659 face images of 15 participants recorded over several months during everyday laptop use [9]. The face-based variant MPIIFaceGaze is commonly evaluated with a leave-one-subject-out protocol.
- **GazeCapture** [1] contains 2 445 504 crowdsourced frames from 1474 participants on Apple mobile devices (iPhones and iPads), with large variation in illumination and head pose. GazeCapture forms the data foundation of the baseline experiments in this work.
- **ETH-XGaze** [10] is a high-resolution multi-camera dataset with more than one million images of 110 participants and extreme variations in head pose and gaze angle.
- **EVE** [11] combines time-synchronized screen recordings, user-facing camera views, and gaze data. The dataset was collected from 54 participants and comprises four camera views with more than 12 million frames in total.
- **Gaze360** [12] contains indoor and outdoor recordings of 238 participants with 3D gaze labels across a very wide range of head poses and distances (about 172 000 frames in the training, validation, and test splits).

- **EyeDiap** and **RT-GENE** are mainly used in the literature for cross-dataset evaluations; RT-GENE includes, among other things, larger camera distances and head-mounted devices.

Table 2.1: Overview of selected gaze datasets.

Dataset	Size	Participants	Recording conditions
MPIIGaze	213 659 images	15	Laptop webcam, everyday use over months
GazeCapture	2 445 504 frames	1474	Smartphones and tablets, crowdsourcing
ETH-XGaze	>1 million images	110	Multi-camera laboratory, extreme head poses
EVE	>12 million frames	54	Screen-camera setup, video sequences
Gaze360	approx. 172 000 frames	238	Indoor and outdoor, wide head-pose and distance ranges

2.4 Deep Learning Architectures for Gaze Regression

2.4.1 Convolutional Neural Networks

CNNs such as ResNet-18/50 [2], VGG, or MobileNet are the standard feature extractors for eye and face crops. Through their local convolution operations, they capture fine textures and edge gradients of the eye region. Specialized gaze models also frequently build on CNNs: FAZE [13] uses a DenseNet-based encoder-decoder to disentangle gaze, head pose, and appearance. In the present work, ResNet18 serves as the convolutional reference model.

2.4.2 Vision Transformers

Vision Transformers (ViTs) [14] split an image into patches and use self-attention to model global spatial dependencies, for example between facial features, head orientation, and the eye region. UniGaze [15] establishes MAE-pretrained ViTs of the sizes Base, Large, and Huge as powerful backbones for gaze estimation. TTP-GE [16], in contrast, combines two ResNet-18 streams with a Transformer-based fusion into a hybrid architecture. GazeCLIP [17] shows that Transformers are not inherently superior: there, a CLIP-pretrained ResNet-50 achieves 3.5° on MPIIFaceGaze, whereas the ViT-B/32 variant achieves 3.7°. Ye et al. [18], in turn, show with *TransGaze* that "plain" ViTs with large-scale pretraining can achieve very good results on GazeCapture and MPIIFaceGaze; among the pretraining strategies examined, self-supervised DINO pretraining performs best. These results motivate evaluating convolutional

and Transformer-based encoders alike rather than assuming one architecture to be universally superior.

2.4.3 Pretrained Foundation Backbones

Transfer learning from large-scale visual pretraining provides general-purpose image representations that can be used as frozen feature extractors for downstream regression heads. Four pretraining paradigms are relevant to this work:

- **Supervised ImageNet pretraining**, the classical starting point for CNN backbones such as ResNet [2].
- **DINOv2** [19], a self-supervised method that learns robust visual features without manual annotations.
- **CLIP** [20], which jointly aligns image and text representations through contrastive learning on image-text pairs.
- **Masked Autoencoders (MAE)** [21], which learn representations by reconstructing masked image regions.

How these representations can be used for gaze estimation is discussed in the following section.

2.5 Pretrained Vision Models for Gaze Estimation

2.5.1 Frozen Backbone vs. Full Fine-Tuning

Two fundamental strategies can be distinguished for adapting pretrained models to gaze estimation. In *full fine-tuning*, all backbone weights are updated; this allows maximal adaptation to the target task but requires large amounts of data and, on small datasets, risks overfitting and *catastrophic forgetting*. With a *frozen backbone* (e.g., ResNet-18 or ViT-B/14), the pretrained weights remain unchanged and only a lightweight regression head is trained. This drastically reduces the number of trainable parameters and preserves the general visual representations – an important prerequisite for personalization with little calibration data.

Kumar et al. [22] provide a theoretical and empirical justification for this trade-off. They compare full fine-tuning with *linear probing*, in which only the last linear layer is trained. On ten datasets with distribution shift, fine-tuning achieves on average 2 % higher accuracy in distribution but 7 % lower accuracy out of distribution. The authors attribute this to the pretrained features being distorted when head and backbone are trained simultaneously

while the head is still randomly or poorly initialized. As a remedy, they propose the two-stage strategy LP-FT (linear probing first, then fine-tuning), which combines the benefits of both. This finding is directly relevant to personalization: the calibration data of a single person are small and cover the later usage conditions only partially, so a frozen backbone can protect generalization.

The regression head typically consists of a multi-layer perceptron (MLP) that maps latent embeddings, for example 512-dimensional ResNet feature vectors, to 2D coordinates (x, y). Key design decisions concern the output activation function (e.g., sigmoid, Gaussian, or ReLU) and the loss function (e.g., L1, Smooth L1, or distance-weighted variants). Between the two extremes lie parameter-efficient methods such as adapters or LoRA [23], which are discussed in Section 2.6.3.

2.5.2 Large-Scale MAE Pretraining: UniGaze

Qin et al. [15] present UniGaze, a large-scale pretraining framework specifically for gaze estimation. Using masked autoencoders, visual representations are learned from about 1.6 million normalized face images of more than 260 000 identities. The pretraining corpus combines real sources (VGGFace2, VFHQ, CelebV-Text) with synthetic datasets (FaceSynthetics, SFHQ-T2I, FFHQ-NV, XGaze-Dense). A central finding is the importance of data normalization, which maps face images and gaze labels into a normalized space through rotation and scaling and thereby cancels out most of the geometric variability caused by head pose and distance. Zhang et al. [24] were the first to study this procedure systematically and show that a modified formulation without the scaling factor improves accuracy by 9.5–32.7 %, depending on the evaluation setting. This finding is confirmed for UniGaze: according to the reported ablations (UniGaze-B, real pretraining data only, leave-one-dataset-out), normalization reduces the angular error on XTest from 14.56° to 11.95°; with additional synthetic data, 11.78° is reached.

UniGaze uses ViT backbones of the sizes Base, Large, and Huge with a masking ratio of 75 %. The ViT-Huge configuration is pretrained for 300 epochs, which requires about 120 h on four NVIDIA H100 GPUs. For gaze estimation, the models are then fully fine-tuned with an L1 regression loss on MPIIFaceGaze, ETH-XGaze, EYEDIAP, Gaze360, and GazeCapture. The authors evaluate several protocols:

- **Domain generalization** (training on ETH-XGaze or Gaze360, testing on MPIIFaceGaze or the screen-target subset of EYEDIAP): UniGaze-H achieves 5.57° (X→M), 4.65° (X→ECS), 5.65° (G360→M), and 5.86° (G360→ECS), thus outperforming the compared domain generalization methods, e.g., CLIP-Gaze with 6.41°, 7.51°, 6.89°, and 7.06°.

- **Leave-one-dataset-out** (training on four datasets, testing on the fifth): UniGaze-H achieves 11.29° (XTest), 4.51° (MPIIFaceGaze), 5.47° (GazeCapture), 5.88° (EYEDIAP), and 12.37° (Gaze360).
- **Joint-dataset training** (training on all five training splits): UniGaze-H achieves 4.46° (XTest), 5.08° (MPIIFaceGaze), 3.20° (GazeCapture), 5.16° (EYEDIAP), and 9.07° (Gaze360).

The ablation studies show that more pretraining data improve generalization and that combining real and synthetic data performs better than real data alone. Normalization and sufficient pose diversity are also crucial: when the head-pose diversity of the real pretraining data is restricted, the error on XTest increases from 11.95° to 16.70°. Comparisons with DINO and FaRL suggest that the composition of the pretraining data can be more important than its sheer size.

Limitations. Despite its strong generalization results, UniGaze leaves several questions relevant to webcam-based personalization open. The evaluation relies on training with complete datasets and full fine-tuning rather than on few-shot personalization with a short calibration. Parameter-efficient strategies such as head-only training, LoRA, or adapters are not the focus, and the influence of label noise in short user-specific calibration sessions is not examined. Moreover, pretraining requires substantial computational resources (ViT-Huge: about 120 h on 4×H100). To what extent the improvements transfer to unconstrained webcam conditions with eyeglasses, background changes, low resolution, and operating-system-dependent acquisition differences remains open.

Relevance to This Work. UniGaze is a central reference for gaze-specific pretraining. While UniGaze addresses generalization at the population level, the present work investigates the complementary question of how effectively pretrained representations can be adapted to individual users with about 90 s to 120 s of calibration data (about 2700–3600 frames at 30 fps). UniGaze-pretrained ViTs are evaluated alongside DINOv2 and CLIP representations (Table 2.2).

2.5.3 Vision-Language Models: CLIP-Gaze and GazeCLIP

Two recent works use the pretrained vision-language model CLIP [20] for gaze estimation, but pursue different goals.

Table 2.2: Comparison of UniGaze and the personalization setting of this work.

Criterion	UniGaze-H	UniGaze-B	This work
Pretraining data	1.6 million face images, >260 000 identities	Same corpus	UniGaze weights, DINOv2, CLIP
Backbone	ViT-Huge	ViT-Base	Comparison of pretrained backbones
Fine-tuning	Full model	Full model	Head-only; LoRA/adapters where supported
Calibration data	Complete training datasets	Complete training datasets	90 s to 120 s, approx. 2700–3600 frames
Evaluation	Cross-dataset, domain generalization	Cross-dataset, domain generalization	Subject-disjoint, personalized webcam evaluation
Label noise	No phase-based filtering	No phase-based filtering	Explicit `WAIT`/`MOVE` logging
Computational cost	Approx. 120 h on 4×H100	Reported only for ViT-H	No additional pretraining
Angular error (leave-one-dataset-out)	4.51°–12.37°	4.73°–12.41°	To be determined experimentally

CLIP-Gaze. Yin et al. [25] address domain generalization with CLIP-Gaze, that is, maintaining performance in a target domain without labeled target data. The core idea is to model gaze-irrelevant factors through text descriptions of the form "An image of a face with {condition}." These factors are organized into three groups: appearance (identity, facial expression, hair, age, etc.), wearable accessories (eyeglasses, hat, make-up, etc.), and image quality (noise, blur, illumination). A distillation loss aligns the gaze features with the CLIP image encoder, a feature-separation loss reduces the similarity to gaze-irrelevant features, and a feature-rank loss ensures that the similarity ordering of the features matches the ordering of the gaze labels. In addition, *Personalized Context Optimization* (PCO) adapts the text prompts in an identity-dependent manner. The CLIP backbone remains frozen; a separate gaze estimation network consisting of ResNet-18, an MLP feature filter, and a regression head is trained. With ETH-XGaze and Gaze360 as source datasets and MPIIFaceGaze and EyeDiap as target datasets, CLIP-Gaze achieves a mean angular error of about 6.97° across four cross-dataset tasks.

GazeCLIP. Wang et al. [17], in contrast, integrate coarse semantic gaze information into supervised gaze regression. A prompt module determines a coarse gaze direction from {Front, Down, Left, Right} via zero-shot classification with prompts of the form "A photo of a face gazing [class]". A CLIP-pretrained ResNet-50 serves as the fine-tuned image

encoder, while the 12-layer Transformer text encoder remains frozen. A cross-attention module with text as query and image features as key and value fuses both modalities; a three-layer MLP ($1024 \to 256 \to 128 \to 2$) predicts pitch and yaw. GazeCLIP achieves 3.5° on MPIIFaceGaze, 4.7° on EyeDiap, and 7.3° on RT-GENE. The ablations show that cross-attention (3.6°) outperforms concatenation (4.2°) and summation (4.4°), and that combining a fine-tuned image encoder with a frozen text encoder (3.6°) is considerably better than two frozen encoders (8.9°). Reported failure cases include poor illumination and occlusions by hair or eyeglasses; moreover, the zero-shot classifier may produce incorrect coarse directions under domain shift.

Limitations and Relevance to This Work. Both works show that CLIP representations are usable for gaze estimation, but neither investigates systematic personalization with a short calibration. CLIP-Gaze leaves open whether fine-tuning or parameter-efficient adaptation of the CLIP image encoder improves person-specific estimation. GazeCLIP relies on full supervised training with benchmark data, addresses neither few-shot personalization nor the filtering of erroneous calibration labels, and its data are available only upon request according to the publication. The present work therefore investigates whether CLIP features offer advantages over DINOv2- and MAE-based representations under a controlled webcam personalization protocol with head-only training, LoRA, or adapters (Table 2.3). The reported angular errors serve as benchmark references and are not directly comparable to the screen-diagonal-normalized error used in this work.

Table 2.3: Comparison of CLIP-Gaze, GazeCLIP, and the planned CLIP personalization experiments. The right column describes an experimental plan, not a completed result.

Aspect	**CLIP-Gaze [25]**	**GazeCLIP [17]**	**Planned experiments**
Goal	Domain generalization	Supervised gaze regression	Few-shot webcam personalization
Use of CLIP	Image and text encoders frozen	Image encoder fine-tuned, text encoder frozen	Image encoder: frozen or PEFT
Architecture	ResNet-18, MLP filter, regression head	CLIP ResNet-50, cross-attention, MLP	CLIP model compared with DINOv2, MAE, ResNet18
Target-domain labels	None (standard setting)	Complete training data	90 s to 120 s calibration
Reported error	Approx. 6.97° (mean over 4 tasks)	3.5° / 4.7° / 7.3°	To be determined experimentally
Data quality	Modeling of gaze-irrelevant factors	Prompt-based coarse direction (uncertain)	`WAIT` frames are excluded

2.6 Data-Efficient Personalization

Subject-independent models often exhibit a systematic offset for new users, which stems from individual differences in eye anatomy, facial geometry, and head pose. Park et al. [13] refer in this context to inter-personal anatomical differences that limit the accuracy of person-independent networks. Personalization aims to close this gap with as little calibration effort as possible. The following subsections discuss classical calibration schemes, few-shot and meta-learning approaches, and parameter-efficient adaptation methods.

2.6.1 Calibration Methods

In active calibration, the person successively fixates known target points on the screen, whose coordinates serve as labels for person-specific adaptation. Typical schemes use a few fixed points or continuous trajectories across the screen. iTracker [1] uses up to 13 fixed calibration locations and trains a user-specific SVR on them (Section 2.1.2). He et al. [26] show that suitable personalization methods achieve better results with only 2–5 calibration points per person than earlier methods requiring more than 13 points. On smartphones, their best personalized model achieves a 24–26 % lower mean error than the state of the art at that time with at most five calibration points and runs entirely on the device in real time. In addition, they propose an unsupervised personalization method that uses only unlabeled face images. Valliappan et al. [7] use, besides a dot calibration, a *zig-zag calibration* in which a target moves slowly from the top left to the bottom right over 60 seconds while the front camera records at 30 Hz. With about 100 calibration frames, the error can already be reduced almost fourfold.

Beyond the question of *how much* calibration data is needed, the methods differ in *which part* of the model is adapted:

- **Bias or offset calibration:** Chen and Shi [27] decompose the gaze angle into a subject-independent term and a subject-dependent bias. They base this on the observation that the inter-subject squared bias exceeds the intra-subject variance. Calibration then requires only a single gaze point, from whose recordings the bias is estimated. On MPIIGaze and EYEDIAP, the method outperforms comparable single models by up to 10 %; about nine calibration images already reduce the error considerably.
- **Low-dimensional person parameters:** Lindén et al. [28] model personal variations with the *Spatial Adaptive Gaze Estimator* (SPAZE) as a low-dimensional latent parameter space. Calibration reduces to a small optimization problem and is therefore hardly prone to overfitting. With nine calibration samples, SPAZE achieves 2.70° on MPIIGaze, improving on the state of the art at that time by 14 %. The authors also

show empirically that personal variations are well described by three latent parameters per eye.

- **Differential estimation:** Liu et al. [29] train a differential CNN that predicts the gaze difference between two eye images of the same person. Together with labeled reference images from the calibration, this yields the gaze for a new image. Since both images stem from the same person, nuisance factors such as alignment, eyelid closure, or illumination partly cancel out. On EYEDIAP, MPIIGaze, and UT-Multiview, the method outperforms the state of the art even with a single calibration sample.
- **Calibration-free personalization:** Zhao et al. [30] propose the *Episode-based Personalization Network* (EbPN), which aims to compensate for person-specific biases without calibration data through a separate personalization branch.

Taken together, these works show that a large part of the person-specific error can be explained by a few, specifically adapted parameters. This supports the strategy pursued in the present work of adapting only a lightweight regression head on top of a frozen backbone.

Cheng et al. [3] point out a fundamental trade-off: longer calibrations lead to more accurate estimates but impair usability. High accuracy with fast and simple calibration is therefore considered an open research direction. Another frequently neglected aspect is the quality of the calibration labels themselves: when a target jumps, the gaze follows with a physiological delay, so that frames in the transition phase may be mislabeled (Section 2.9).

2.6.2 Few-Shot Learning and Meta-Learning: FAZE

Park et al. [13] present FAZE (*Few-Shot Adaptive Gaze Estimation*), an influential approach to person-specific gaze estimation with few calibration samples. The method combines disentangled gaze representations with Model-Agnostic Meta-Learning (MAML). Instead of training a separate model from scratch for each person, FAZE learns an initialization that can be adapted quickly with a few labeled samples.

FAZE consists of two components. A DenseNet-based *Disentangling Transforming Encoder-Decoder* (DT-ED) represents appearance, gaze direction, and head pose in separate latent codes. Rotation-equivariant representations allow changes in gaze and head orientation to be modeled as transformations in the latent space. Training on image pairs of the same person combines a reconstruction loss, a gaze direction loss, and an embedding consistency term. The second component is a gaze estimator meta-trained with MAML: during meta-training, an initialization is learned across many training subjects from which a few person-specific gradient steps produce a personalized estimator. This is intended to reduce the risk of overfitting associated with conventional fine-tuning on very small datasets.

FAZE is trained on GazeCapture and evaluated on GazeCapture and MPIIGaze. With $k = 9$ calibration samples, the method achieves a mean angular error of 3.14° on MPIIGaze and consistently improves on the compared methods for $1 \leq k \leq 256$. On GazeCapture, the error decreases by 19 % relative to the state of the art, from 3.91° to 3.18°.

Limitations and Distinction. FAZE shows that the effectiveness of calibration depends not only on the amount of data but also on the previously learned representation and the adaptation strategy. However, its representation is based on supervised training with gaze data and a DenseNet architecture, not on general self-supervised (DINOv2, MAE) or multimodal (CLIP) representations. Adaptation is performed via meta-learning; simpler parameter-efficient alternatives such as head-only training, LoRA, or adapters are not investigated as a central strategy. Webcam-specific conditions such as varying image quality, eyeglasses, backgrounds, and operating systems are not in focus either, nor is explicit filtering of calibration frames based on the acquisition phase. The present work addresses these points. Since FAZE reports angular errors, whereas this work uses an error normalized by the screen diagonal, the figures are not directly comparable.

Table 2.4: Comparison of FAZE and the present work.

Aspect	**FAZE [13]**	**This work**
Goal	Few-shot personalization of gaze direction	Data-efficient personalization of the screen position via webcam
Representation	Disentangled codes, DenseNet encoder-decoder	ResNet18 reference and DINOv2-, CLIP-, and MAE-based representations
Pretraining	Supervised on GazeCapture	Pretrained models with different objectives and data sources
Adaptation	MAML and person-specific gradient steps	Head-only training; LoRA and adapters as alternatives
Calibration data	$k = 1$ to 256 samples	Approx. 90 s to 120 s (approx. 2700–3600 frames)
Label quality	No phase-based filtering	Exclusion of `WAIT` frames via phase logging
Evaluation	GazeCapture and MPIIGaze	Multiple webcam sessions under varying conditions
Reported error	3.14° (MPIIGaze, $k = 9$)	Normalized diagonal error, not directly comparable

In addition, Yu et al. [31] extend few-shot adaptation by generating additional training samples from a few reference images of a person via gaze redirection synthesis and using them to fine-tune a generic model.

2.6.3 Parameter-Efficient Fine-Tuning (PEFT)

Parameter-efficient methods adapt only a small fraction of the model parameters and leave the pretrained backbone largely unchanged. This reduces memory requirements, computational cost, and the risk of overfitting when calibration data are scarce. Three variants are relevant to the present work:

- **Head-only training (FC-only):** The backbone remains completely frozen; only the regression head is trained. Since the backbone features can be precomputed and cached, this method is particularly efficient.
- **Adapters** [32]: Small additional modules are inserted between the layers of a frozen network and trained as the only components.
- **Low-Rank Adaptation (LoRA)** [23]: The pretrained weight matrices W remain frozen; an additive update $\Delta W = BA$ with low-rank matrices of rank r is trained, which greatly reduces the number of trainable parameters.

Such methods have hardly been studied systematically in gaze estimation so far. One exception is TPGaze [33], which transfers the idea of prompt tuning from natural language processing: an additional prompt with less than 1 % of the parameters of a ResNet-18 is adapted at test time without labels, with meta-learning ensuring that the unsupervised optimization actually reduces the gaze error (Section 2.8). Bao and Lu [34] further show that precisely the fully connected output layer, which maps high-dimensional image features to gaze, is highly prone to overfitting. Consequently, the design of the regression head matters even for head-only training. The remaining works considered use either full fine-tuning (UniGaze, GazeCLIP), meta-learning (FAZE), or a frozen pretrained model to guide a separately trained network (CLIP-Gaze). The present work therefore investigates how well head-only training and further parameter-efficient methods are suited for personalizing pretrained models with short webcam calibrations.

2.7 Domain Shift, Generalization, and Evaluation Protocols

2.7.1 Subject-Independent Evaluation

How well a gaze estimator generalizes to new people depends crucially on the evaluation protocol. With a random split of individual frames (`norm_random`), images of the same person can appear in both the training and the test split. The model can then exploit person-specific features, and generalization is overestimated. Subject-disjoint splits (`norm_subject`), for example in the form of a leave-one-subject-out protocol, rule out this data leakage and measure performance on truly unseen people. In the baseline experiments of this work on

GazeCapture, `norm_subject` yields a test error of 20.04 %, whereas `norm_random` yields 18.87 % (see Chapter 4).

These must be distinguished from cross-dataset protocols such as the leave-one-dataset-out evaluation in UniGaze [15], in which an entire dataset is held out. Subject-disjoint and cross-dataset evaluations thus measure complementary forms of generalization: to unseen people within a domain and to unseen recording conditions, respectively.

2.7.2 Domain Adaptation and Domain Generalization

When training and test data differ in people, cameras, or environments, the accuracy of appearance-based methods drops considerably. Two strategies address this domain shift: domain adaptation uses (mostly unlabeled) data from the target domain, whereas domain generalization requires no target data at all.

CRGA. Wang et al. [35] propose *Contrastive Regression Gaze Adaptation* (CRGA), a method for unsupervised domain adaptation. A *Contrastive Domain Generalization* (CDG) module learns stable representations in the source domain by pulling together features with similar gaze labels and pushing apart features with strongly differing labels. A *Contrastive Self-training Adaptation* (CSA) module then learns from pseudo-labels in the target domain. With ETH-XGaze and Gaze360 as source domains and MPIIGaze, RT-GENE, GazeCapture, and EyeDiap as target domains, the authors report improvements over the baseline of 40.4 %, 34.7 %, 55.8 %, and 34.3 % (source ETH-XGaze) and 31.7 %, 30.5 %, 32.9 %, and 23.8 % (source Gaze360), respectively.

CRGA thus addresses a complementary problem: adapting to a new domain without labeled target data. The present work, in contrast, uses a small amount of labeled calibration data from the target person, which enables supervised fine-tuning with real labels instead of pseudo-labels. Self-training with pseudo-labels also carries the risk of confirmation bias when erroneous predictions are reinforced during further training. Moreover, this work focuses on person-specific variations rather than transfer between benchmark datasets. CRGA should therefore be regarded as related work on robustness to domain shift, not as a directly comparable baseline.

Domain Generalization. Domain generalization methods such as PureGaze [36] or CLIP-Gaze [25] (Section 2.5.3) attempt to remove gaze-irrelevant factors from the features already during training, so that the model transfers to new domains without target data. UniGaze [15] shows that large-scale gaze-specific pretraining can outperform these methods in the typical cross-dataset setting (Section 2.5.2).

2.7.3 Multi-Person Gaze Estimation

Zhang et al. [37] present GazeOnce, a one-stage method that estimates the gaze directions of all people in an image in a single pass and remains real-time capable even with many faces (more than ten per image). In addition to the gaze direction, auxiliary face information such as bounding boxes and landmarks is predicted. A projection-based self-supervised loss promotes consistency between the estimated 3D gaze direction and its 2D projections. For training, the authors introduce the synthetic dataset MPSGaze, which combines faces from WiderFace with gaze labels from ETH-XGaze: for each suitable WiderFace face, a matching ETH-XGaze face is retrieved and swapped in. According to the authors, GazeOnce outperforms existing methods in both accuracy and speed, especially with many faces.

GazeOnce thus addresses a different problem from this work: the simultaneous gaze estimation of multiple people in unconstrained scenes rather than personalization for a single webcam user. Methodologically relevant is the projection-based consistency constraint. Moreover, MPSGaze illustrates that synthetically generated training data entail potential domain differences from real webcam recordings.

2.7.4 Device and Environmental Variations

Beyond differences between datasets, numerous device- and environment-related factors affect the accuracy of webcam-based systems. These include eyeglasses and occlusions, background and illumination, image resolution and camera distance, and the geometry between camera and screen. The literature provides several indications:

- Cheng et al. [3] observe considerably larger errors on datasets with large distances and low resolution (Gaze360, RT-GENE).
- GazeCLIP [17] names poor illumination and occlusions by hair or eyeglasses as typical failure cases.
- UniGaze [15] highlights the importance of head-pose diversity and the range of gaze labels.
- CLIP-Gaze [25] explicitly models accessories and image quality as gaze-irrelevant factors.

However, systematic investigations of these factors in the context of personalization with short webcam calibrations are largely lacking. The present work therefore additionally considers acquisition differences between operating systems (Ubuntu and Windows), eyeglasses, backgrounds, and calibration durations across multiple recording sessions (see Chapter 4).

2.8 Test-Time Adaptation

Test-time adaptation (TTA) refers to methods that adapt an already trained model to the target person or target environment only during use, typically with few or no labels at all. Unlike classical calibration (Section 2.6.1), adaptation should take place with as little active involvement of the user as possible.

2.8.1 TTP-GE: Self-Supervised and Weakly Supervised Adaptation

Nagpure et al. [16] present TTP-GE (*Test-Time Personalized Gaze Estimation*), a two-stage adaptation method for in-vehicle gaze estimation. It builds on GazeDPTR V2, an architecture with two independent processing streams for the original and an affine-normalized face image, whose features are fused by a Transformer. Positional information from a point projection enables gaze zone classification in addition to gaze regression.

Meta-training follows a two-level optimization scheme based on MAML. In the inner loop, a self-supervised consistency loss in the style of BYOL++ adapts the model to new people without requiring gaze labels. The outer loop optimizes gaze regression and zone classification with labeled training data. At test time, two complementary adaptation steps are performed: self-supervised adaptation on unlabeled images of the target person responds to changed image and recording conditions. Weakly supervised adaptation of the zone classifier then uses a few semantic zone labels (e.g., mirrors, dashboard, windshield) collected during a short setup phase.

On IVGaze (44 705 images of 125 subjects), the combined strategy achieves a mean angular error of 6.00°, compared with 6.78° for the baseline, 6.13° for weakly supervised adaptation, and 6.72° for purely self-supervised adaptation. The proportion of estimates with an error below 6° increases from 49.95 % to 58.2 %. The two adaptation stages are therefore complementary: self-supervision addresses the image distribution, whereas the weak zone labels provide person-specific information.

2.8.2 Further Approaches to Label-Efficient Personalization

Several recent works pursue similar goals with different means:

- **TPGaze** [33] adapts only a meta-learned prompt with less than 1 % of the parameters of a ResNet-18 at test time. A simple symmetry loss suffices as the unsupervised objective. According to the authors, adaptation is up to ten times faster than with the compared methods, and it is evaluated in four cross-dataset settings.

- **ELF-UA** [38] formulates *label-free* user adaptation as a problem in its own right: a few unlabeled images of the target person suffice for adaptation. Meta-learning uses a generalization bound from domain adaptation as its loss function.
- **PIGaze** [39] automatically obtains in-vehicle calibration data from the geometric consistency of gaze on known zones such as mirrors and the windshield (self-calibration). A pretrained baseline model serves as a *frozen* feature extractor, on top of which a lightweight adapter learns the relative gaze mapping. On IVGaze and LBW, PIGaze outperforms fine-tuning approaches.

Limitations and Distinction. The methods above aim to reduce calibration effort through unlabeled or weakly labeled data. They are mostly geared toward 3D gaze directions and, in the case of TTP-GE and PIGaze, toward the in-vehicle domain with fixed semantic zones. Their adaptation procedures (meta-training, online updates, EMA target networks) are also considerably more complex than simple head-only training. General pretrained representations such as DINOv2, CLIP, or MAE are not investigated as backbones. The present work pursues the complementary approach of *supervised* personalization with continuous screen coordinates from a short calibration. Notably, however, PIGaze – like the present work – combines a frozen backbone with a lightweight, person-specifically trained module.

Table 2.5: Comparison of TTP-GE and the present work.

Aspect	**TTP-GE [16]**	**This work**
Paradigm	Test-time adaptation with unlabeled images and weak zone labels	Supervised personalization with labeled calibration data
Backbone	GazeDPTR V2: two ResNet-18 streams with Transformer fusion	ResNet18 reference and DINOv2-, CLIP-, and MAE-based representations
Labels	Unlabeled images and nine semantic gaze zones	Continuous screen coordinates; `WAIT` frames are excluded
Adaptation	Meta-training, online updates, EMA target network	Head-only training; PEFT variants as alternatives
Domain	In-vehicle driver monitoring	Webcam use in front of a screen
Result	6.00° on IVGaze	Normalized diagonal error, not directly comparable

2.9 Data Quality, Label Noise, and Temporal Dynamics

Supervised personalization assumes that each calibration image is labeled with the point at which the person is actually looking at that moment. In practice, this assumption holds only approximately. Discrepancies between label and actual gaze act as *label noise* and are particularly critical for short calibrations, since each erroneous sample constitutes a larger share of the training data.

2.9.1 Data Quality in Eye Tracking

In eye-tracking research, data quality is commonly described by three quantities: *accuracy* (systematic deviation between measured and true gaze point), *precision* (dispersion of repeated measurements), and *data loss*. Niehorster et al. [40] provide a recent overview of how these quantities are operationalized and offer tools for computing them. Learning-based methods add a further dimension: the quality of the *training labels*. Erroneous labels distort not only the evaluation but also the learned model itself.

2.9.2 Reaction Latency and Saccadic Delay

When a calibration target appears at a new position, the gaze does not follow immediately. A physiological latency elapses between the appearance of the target and the onset of the saccade. In a prosaccade task on smartphones, Valliappan et al. [7] measure a mean saccade latency of 210 ms (median 167 ms), consistent with the 200 ms to 250 ms observed in earlier studies. This is followed by the duration of the saccade itself and possibly corrective saccades until the eye fixates stably on the target.

Frames recorded during this transition interval carry the label of the *new* target position but show a gaze still directed at the *old* position or somewhere in between. At a recording rate of 30 fps, a latency of about 200 ms alone affects roughly six frames per target jump. The more frequently the target jumps, the larger the share of systematically mislabeled frames.

2.9.3 Preventive Filtering During Data Acquisition

An obvious countermeasure is to discard frames immediately after a target change. Valliappan et al. [7], for example, discard the first 800 ms after the onset of each stimulus across the board to account for saccade latency. Such a fixed time window is easy to implement but relies on an assumption about the duration of the transition phase and is usually not evaluated as an object of study in its own right.

The present work pursues a related but more explicit approach: the acquisition protocol logs the current *phase* of the calibration target for each frame, namely `WAIT` (a short waiting phase of 0.5 s immediately after a target jump) or `MOVE` (the target moves continuously during the following 4.5 s). Frames of the `WAIT` phase can thus be identified deterministically and reproducibly and excluded before training, which affects about 10 % of the frames in the recorded sessions. Unlike a fixed time window, phase logging makes it possible to study the effect of this exclusion specifically, that is, to compare models trained with and without cleaned data under otherwise identical conditions.

2.9.4 Post-Hoc Filtering and Robust Learning Methods

Preventive methods contrast with methods that detect erroneous labels *after* data acquisition based on statistical or model-based criteria.

DMRF. Jiang et al. [41] propose *Dynamic Multi-model Regression Filtering* (DMRF), a general method for label noise filtering in regression tasks. It addresses a chicken-and-egg problem: noisy labels bias the models used to estimate the noise, and biased models in turn lead to incorrect noise estimates. DMRF combines three components:

1. a *warm-up filtering* based on a distribution function of the estimated errors,
2. a *dynamic model selection* according to a correlation criterion,
3. a *data recall* step that returns samples with small loss to the clean subset.

Besides 15 UCI and KEEL benchmarks and an age estimation task, DMRF is integrated into the gaze estimation method L2CS-Net. On a preprocessed MPIIGaze dataset with 45 000 images and leave-one-person-out evaluation, the mean angular error decreases from 3.92° to 3.85°, with 74.3 % of the data retained. The code is publicly available.

SeeTN. Peng et al. [42] are the first to systematically investigate the influence of label noise on domain generalization in gaze estimation. In a preliminary experiment, 10, 20, and 30 % of the Gaze360 labels are corrupted with strong Gaussian noise; as a result, the cross-dataset error of a ResNet-18 baseline on MPIIGaze increases from 7.94° to 12.17° (20 %) and 14.38° (30 %), respectively. The proposed *See-Through-Noise* framework constructs a semantic embedding space, distinguishes noisy from clean samples based on the consistency between features and labels, and transfers gaze-relevant information from clean to noisy samples.

Preventive vs. Corrective. Post-hoc methods such as DMRF and SeeTN detect erroneous labels from the data themselves and can therefore also capture unanticipated sources of error. However, they require sufficiently many samples for a reliable estimate, are computationally more expensive, and have so far been evaluated on population-level datasets rather than on short person-specific calibrations. Preventive methods such as phase logging, in contrast, capture only a *known*, protocol-related source of error, but do so deterministically and without an additional model. The two approaches are therefore complementary (Table 2.6).

Table 2.6: Comparison of post-hoc filtering (DMRF) and preventive phase logging.

Criterion	**DMRF [41]**	**This work**
Strategy	Corrective: estimation and filtering after acquisition	Preventive: logging of the acquisition phase
Detection	Model predictions, error patterns, dynamic model selection	Explicit `WAIT`/`MOVE` labels
Errors captured	Any statistically conspicuous samples	Only the protocol-related transition phase
Data retained	74.3 % (MPIIGaze)	Approx. 90 %
Setting	Population level, leave-one-person-out	Person-specific webcam calibration
Additional cost	Iterative training of multiple regression models	No additional model
Result	$3.92° \rightarrow 3.85°$	Normalized diagonal error, not directly comparable

2.10 Comparison and Critical Discussion of Existing Approaches

Table 2.7 summarizes the central works of this chapter along the dimensions most relevant to this work: goal and paradigm, representation used, type of personalization, and handling of label quality.

Critical Discussion. Four overarching observations can be derived from the comparison:

1. **Representation and personalization are studied separately.** Works on large-scale pretrained models (UniGaze, CLIP-Gaze, GazeCLIP) aim at generalization across datasets and use full fine-tuning on extensive training data. Works on personalization (FAZE, SPAZE, Chen & Shi), in contrast, rely on task-specifically trained CNNs. How well general pretrained representations can be personalized with little calibration data remains open.

2. **Few parameters often suffice.** Bias calibration, low-dimensional person parameters, prompt tuning, and frozen base models with adapters (PIGaze) consistently show

Table 2.7: Systematic comparison of central works. The reported errors stem from different protocols and metrics and are not directly comparable.

Work	**Goal / paradigm**	**Representation**	**Personalization**	**Label quality**	**Result**
iTracker [1]	2D gaze, mobile	AlexNet-like CNN streams	SVR, up to 13 points	–	1.34 cm (phone)
FAZE [13]	Few-shot	DenseNet DT-ED, supervised	MAML, $k \leq 256$	–	3.14° ($k = 9$)
SPAZE [28]	Few-shot	CNN, latent person parameters	Small optimization problem	–	2.70° ($k = 9$)
Chen & Shi [27]	Offset calibration	CNN	Bias from a single gaze point	–	up to 10 % better
Valliappan [7]	Smartphone tracking	CNN, pretrained on GazeCapture	Fine-tuning + regression, approx. 30 s	Fixed 800-ms discard	0.46 cm
CRGA [35]	Unsupervised DA	ResNet (ImageNet)	Pseudo-labels	–	up to 55.8 % better
UniGaze [15]	Generalization	MAE-ViT, 1.6 million faces	Full fine-tuning	–	4.65–5.86° (DG)
CLIP-Gaze [25]	Generalization	CLIP (frozen) + ResNet-18	–	Gaze-irrelevant factors	6.97° (DG)
GazeCLIP [17]	Supervised regression	CLIP ResNet-50 (fine-tuned)	–	–	3.5° (MPI-IFaceGaze)
TTP-GE [16]	Test-time adaptation	GazeDPTR V2	Self- + weakly supervised	–	6.00° (IVGaze)
TPGaze [33]	Test-time adaptation	ResNet-18	Prompt (<1 % params)	–	–
DMRF [41]	Label noise filtering	L2CS-Net	–	Post hoc, model-based	3.85° (MPIIGaze)
SeeTN [42]	Robust generalization	ResNet-18/50	–	Post hoc, feature-based	–
This work	Webcam personalization	ResNet18, DINOv2, CLIP, MAE	Head-only, PEFT; 90 s to 120 s	Preventive, phase logging	Normalized diagonal error

that a large part of the person-specific error can be explained by a small number of adapted parameters. Kumar et al. [22] provide a general justification for this.

3. **Label quality is mostly handled after the fact.** DMRF and SeeTN filter or reweight noisy labels on population-level datasets. Protocol-related errors in short calibrations are handled, if at all, via fixed time windows (Valliappan et al.) without systematically investigating their influence.

4. **The results are hardly comparable.** The works report angular errors in degrees, distances in centimeters, or relative improvements and use different datasets, splits, and protocols. An angular error can be converted into a screen error only for a known geometry (distance, screen size). The metric used in this work, the Euclidean error normalized by the screen diagonal, is therefore not directly comparable with the values in the literature. The figures in Table 2.7 serve for orientation, not as a ranking.

2.11 Research Gaps and Transition to the Present Work

2.11.1 Identified Research Gaps

The analysis of the state of the art reveals four research gaps that the present work addresses:

G1: Data-efficient personalization of general pretrained models. Large-scale pretrained representations (DINOv2, CLIP, MAE or UniGaze) have so far been investigated for gaze estimation mainly with full fine-tuning on extensive datasets. Systematic comparisons of these representations in a personalization setting with only a few minutes of webcam calibration are lacking.

G2: Parameter-efficient adaptation for personalization. Although several works show that a few adapted parameters capture most of the person-specific error, head-only training, adapters, and LoRA have hardly been compared systematically as personalization strategies in gaze estimation.

G3: Label quality of short calibrations. Saccade latency and target changes lead to systematically mislabeled frames in calibration data. Existing works mostly handle label noise post hoc at the population level or discard fixed time windows. How an explicit, protocol-based exclusion of such frames affects personalization has not yet been investigated systematically.

G4: Realistic evaluation under webcam conditions. Many results are based on benchmarks with controlled conditions or on frame-wise random splits that can overestimate generalization. The influence of practical factors such as operating

system, eyeglasses, background, and calibration duration on personalization is hardly documented.

2.11.2 Transition to the Present Work

The present work addresses these gaps with an integrated personalization pipeline for webcam-based eye tracking:

- **Frozen backbone with a lightweight regression head (G1, G2):** Pretrained representations are used as frozen feature extractors; the features are cached, and only a compact MLP regression head is trained person-specifically. ResNet18 serves as the reference and is compared with DINOv2-, CLIP-, and MAE-based representations. Parameter-efficient variants are investigated where the architecture permits.
- **Phase logging and preventive cleaning (G3):** The acquisition protocol logs the `WAIT` and `MOVE` phases of the calibration target. This allows the effect of excluding transition frames to be studied in a controlled manner by comparing models trained on original and cleaned data under otherwise identical conditions.
- **Systematic ablations:** Batch size, learning rate, trainable layers, activation and loss functions (including a distance-weighted L1 variant), and dataset size are examined with repeated runs and confidence intervals.
- **Subject-disjoint and webcam-specific evaluation (G4):** Baseline models are assessed with the subject-disjoint split `norm_subject`. Personalization is evaluated across multiple recording sessions with varying operating system, eyeglasses, background, and calibration duration.

The methodology is described in Chapter 3, the results are presented in Chapter 4, and they are discussed in Chapter 5 with respect to the research questions formulated in Chapter 1.

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